AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 1 · WORKBOOK

Precision in Algorithms

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

One connected first-program evidence record.

Translate the counter calculation, predict and observe outputs, explain an order error and write the independent tile script.

Keep one habit: Keep saved code, expected output, actual output/error and reasoning together.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Expo and Defense. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

▼ Earlier wording, alternatives and picture transcripts

Expo and Defense

MAIN LESSON · SECTION 12

W1 · Translate equal groups into four lines

> Use the supplied complete script reference.

Your evidence record

- Script reference: packs = 4; per_pack = 6; total = packs * per_pack; print(total), with each statement on its own line. The supplied quantities are whole and nonnegative. For W2 change only packs to 5, save and run the complete file; retain expected and actual outputs separately. This script does not yet validate typed input.

- Write the complete four-line script for 4 packs of 6. Beside each line explain what it does. Which line displays information?

My reasoning and evidence

MAIN LESSON · SECTION 13

W2 · Predict before running each saved version

> Record actual outputs or labelled paper traces.

Your evidence record

- Make records for 4 packs of 6, then 5 packs of 6. Predict before running. Record actual **output**, any error and what changed. If working on paper, label both results as traces.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Explain a read-before-assignment error

> Use a fresh run, then restore the valid statement order.

Your evidence record

- Move print(total) to the first line of a fresh script. Predict what happens and why. Put the lines back into an order that meets the requirement.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Keep a checkable, limited conclusion

> Separate execution success from meeting the requirement.

Your evidence record

- Explain the difference between a successful run and a correct calculation. State one limit of this **program** and what you would keep so another person can check your work.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Transfer the procedure independently

> Keep one first attempt before hints or feedback.

Read and explain independently

A fictional shelf has 3 trays of 7 tiles each. Write a four-line Python script using trays, per_tray and tiles. Predict its **output**. Then change only trays to 0 and predict again. Explain why adding the two quantities is wrong.

> **My independent explanation, trace and limits**

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Transfer the procedure independently

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Write exact scoring steps so two people rank the same book the same way.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Predict then observe for From a paper **algorithm** to Python. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for From a paper **algorithm** to Python. Proposed explanation: “An equals sign displays the answer”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

Write an **algorithm** for borrowing a classroom book when the book may already be on loan. Include **input**s, a no-copy-available outcome and a check that prevents two borrowers receiving the same copy.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

A sandwich **algorithm** says add spread, close it, cut it, without stating materials or directions.

Expose **assumption**s and write a **precise** paper simulation.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ Extra practice · A changed case

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

A robot starts at square 0. Replace move a bit and stop nearby with instructions that put it exactly on square 3 using steps of one square.

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 1 Test and reflect

▼ Extra practice · Design a revealing test

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Reference · Original target response space**

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

Write an **algorithm** for borrowing a classroom book when the book may already be on loan. Include **input**s, a no-copy-available outcome and a check that prevents two borrowers receiving the same copy.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

A fictional shelf has 3 trays of 7 tiles each. Write a four-line Python script using trays, per_tray and tiles. Predict its **output**. Then change only trays to 0 and predict again. Explain why adding the two quantities is wrong.

> **My reasoning and evidence**

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

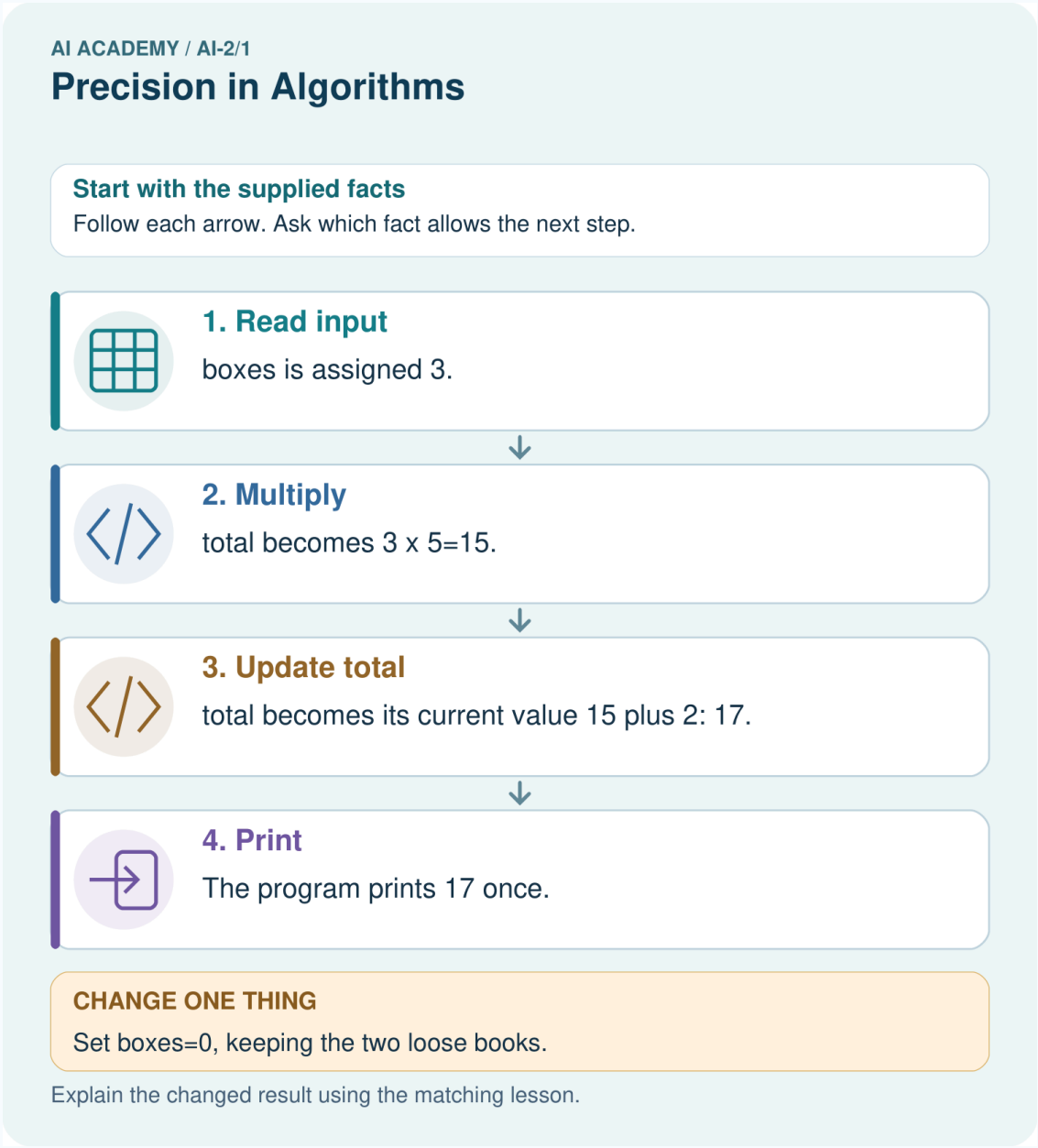
> Keep the explanation and the support you used.

Use this when helpful

A fictional book-box counter has three boxes with five books each and two loose books.

Python instructions: boxes=3; total=boxes*5; total=total+2; print(total).

Run the instructions in this order with ordinary integer arithmetic.



Follow the labelled picture with the matching explanation. Open the vector diagram

If print(total) moves before total=total+2, what gets printed for the original **input**?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Use small predicted traces when translating the Transparent Book Recommender's paper logic into code.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Transparent Book Recommender

Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Use small predicted traces when translating the Transparent Book Recommender's paper logic into code.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional book-box counter has three boxes with five books each and two loose books.
- Which labelled step matches this detail? The **program** prints 17 once.
- 1. Read **input**
- 2. Print
- 3. Multiply

- 4. Update total

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Print The **program** prints 17 once. **Assignment** replaces a variable's stored value.

In total=total+2, Python first reads the current total on the right, calculates 17, then assigns that result to total on the left. The equal sign is an instruction here, not a claim that 15 and 17 are mathematically equal.

Predict **output** before running.

Changed-case question

- Changed-case practice: Set boxes=0, keeping the two loose books. Which conclusion is supported by the supplied facts?

- 1. This one example guarantees correct results for every future **input**.

- 2. The printed result is 2.

- 3. We can decide the result without examining the changed **input** or the procedure.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: The printed result is 2. 0×5+2=2; there is no rule saying zero boxes means zero loose books.

Repair and save your work

Review Expo and Defense (ai-1/36). Goal: Evaluate a saved candidate and a manual baseline fairly. Short practice: Write one defensible success statement and one claim the **evidence** cannot support..

Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

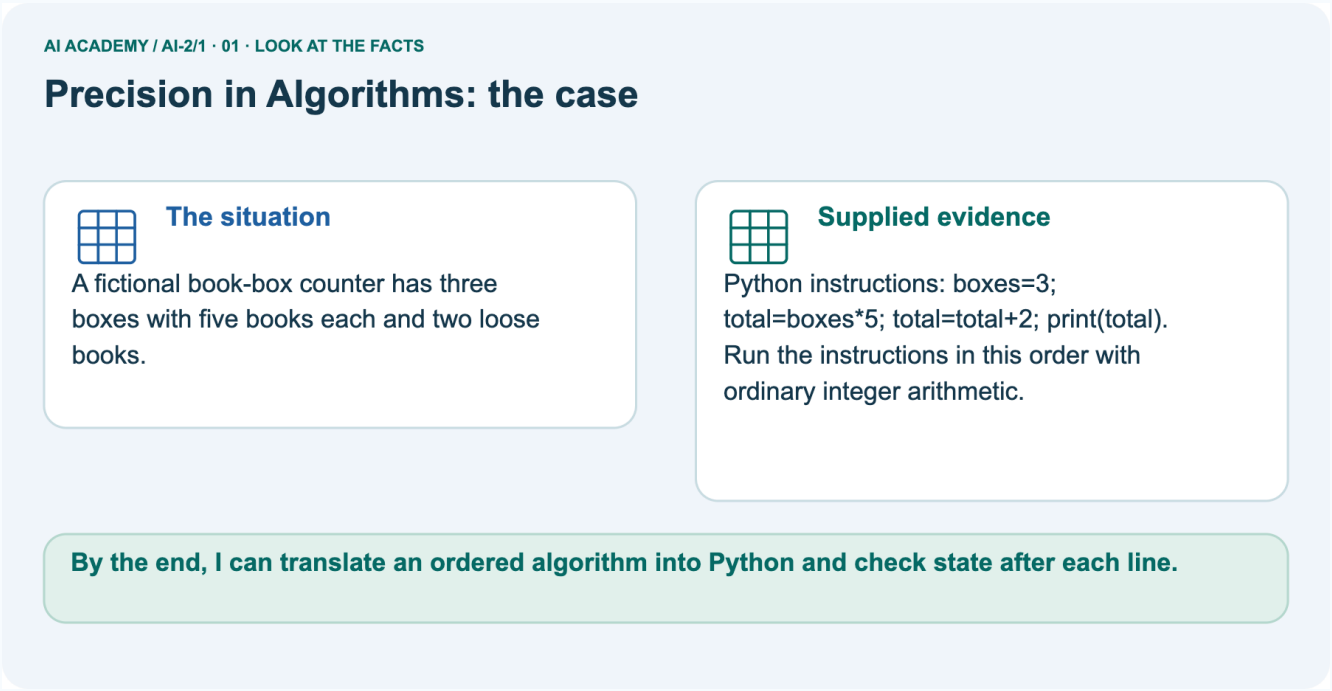
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Precision in Algorithms: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

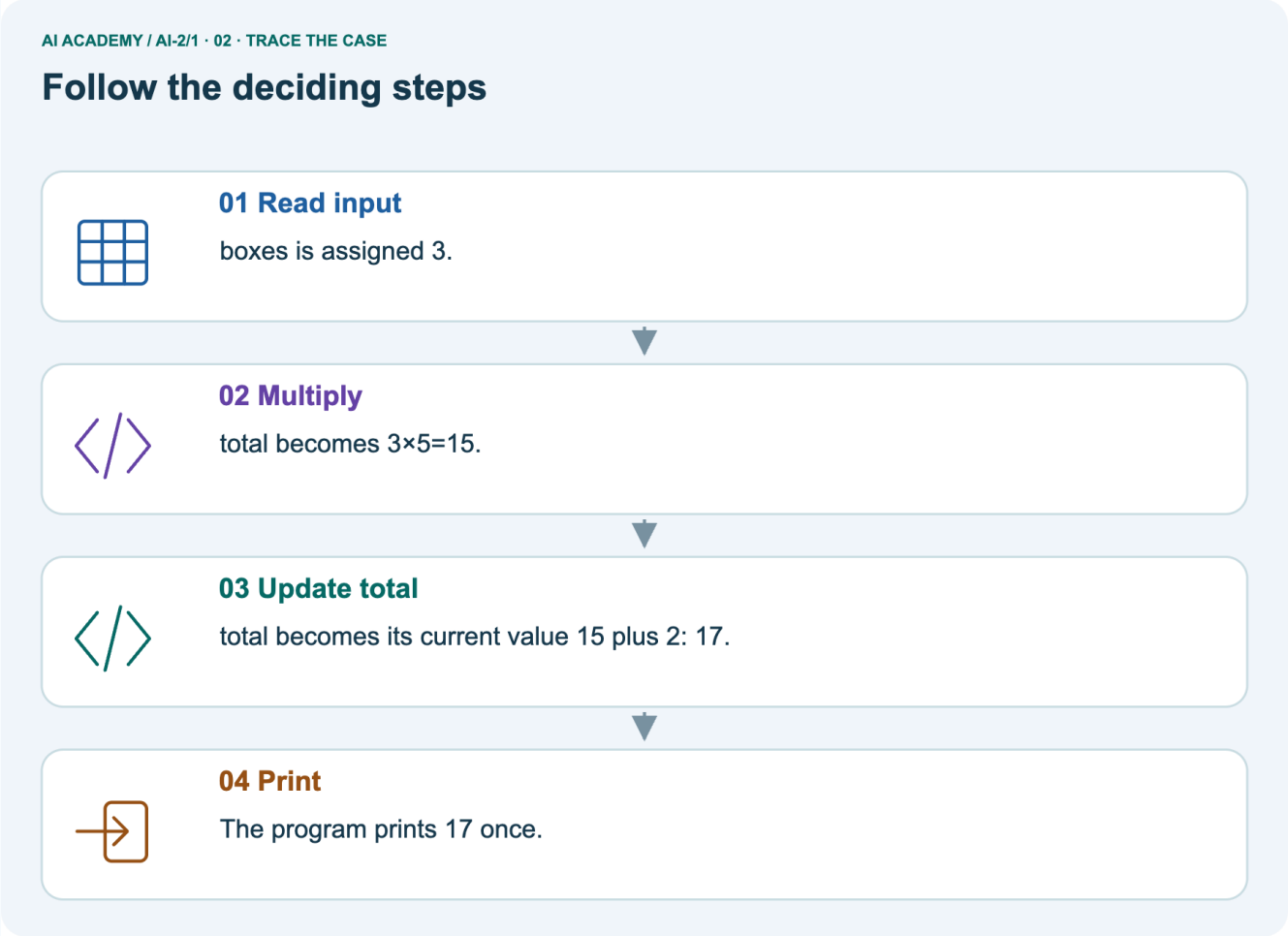
A fictional book-box counter has three boxes with five books each and two loose books.
Supplied facts: Python instructions: boxes=3; total=boxes*5; total=total+2; print(total).

Run the instructions in this order with ordinary integer arithmetic.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Read **input**: boxes is assigned 3. → Multiply: total becomes 3×5=15. → Update total: total becomes its current value 15 plus 2: 17. → Print: The **program** prints 17 once.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. A calculation from paper to output

AI ACADEMY / AI-2/1 · 03 · INSPECT THE LESSON MODEL

A calculation from paper to output

Paper step	Python instruction	Effect
Record pack count	packs = 4	Name holds 4
Multiply two quantities	total = packs * per_pack	Name holds product
Show the answer	print(total)	Displays the value

Assign per_pack before the multiplication. This diagram summarises the steps; the complete script below includes that assignment.

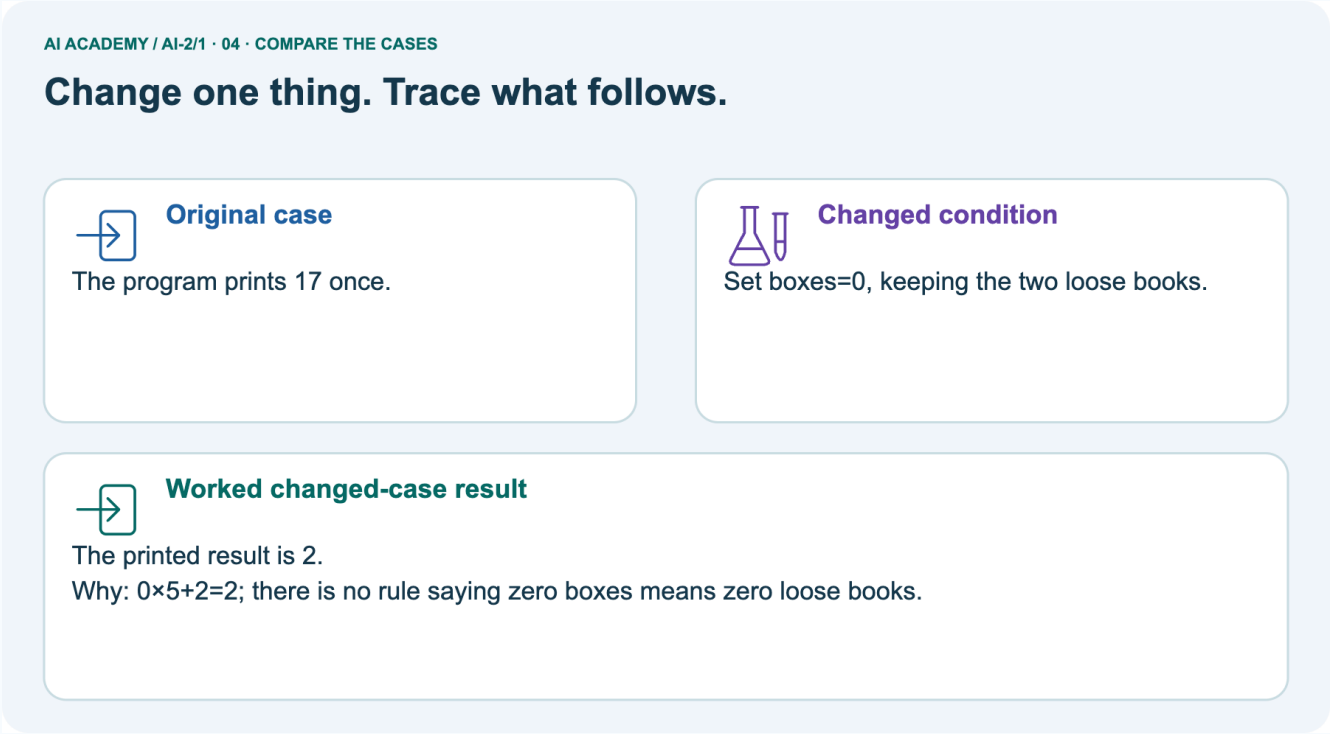
Follow the labelled picture with the matching explanation. Open the vector diagram

A calculation from paper to **output**. Record pack count | packs = 4 | Name holds 4. Multiply two quantities | total = packs * per_pack | Name holds product. Show the answer | print(total) | Displays the value

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



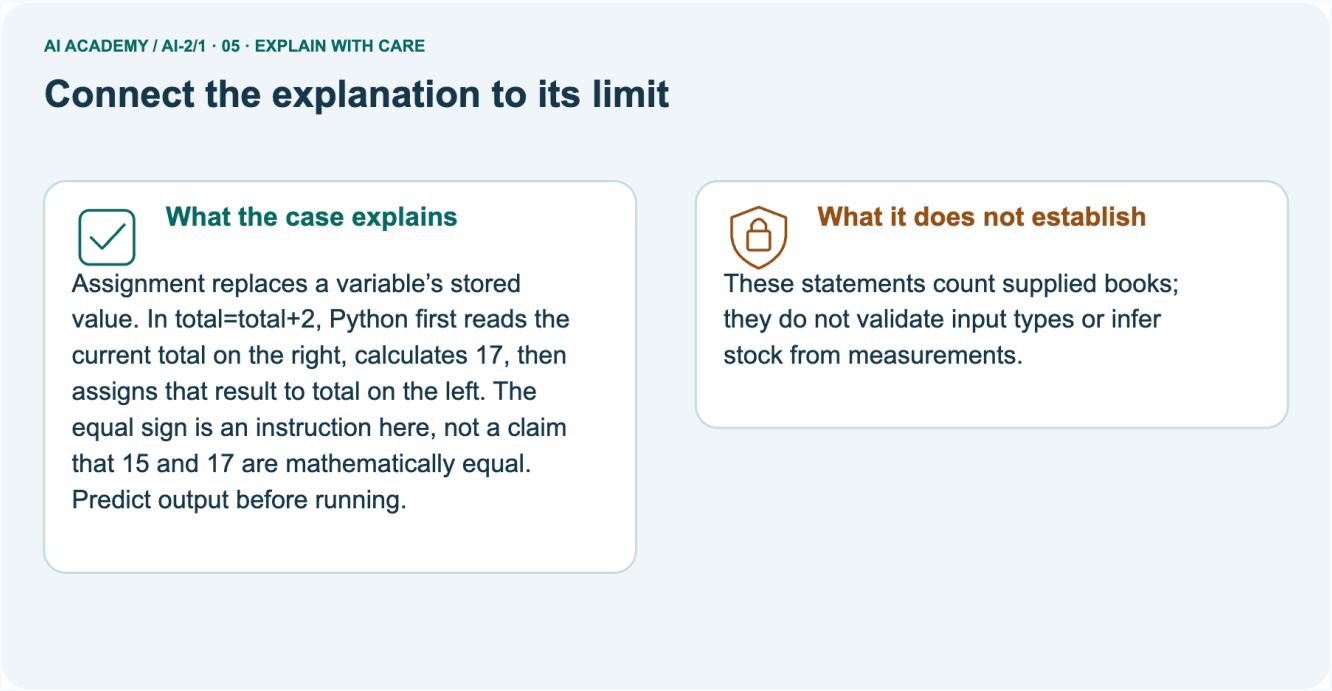
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: The **program** prints 17 once. Changed condition: Set boxes=0, keeping the two loose books. Expected result: The printed result is 2. Reason: 0×5+2=2; there is no rule saying zero boxes means zero loose books.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Assignment replaces a variable's stored value. In total=total+2, Python first reads the current total on the right, calculates 17, then assigns that result to total on the left.

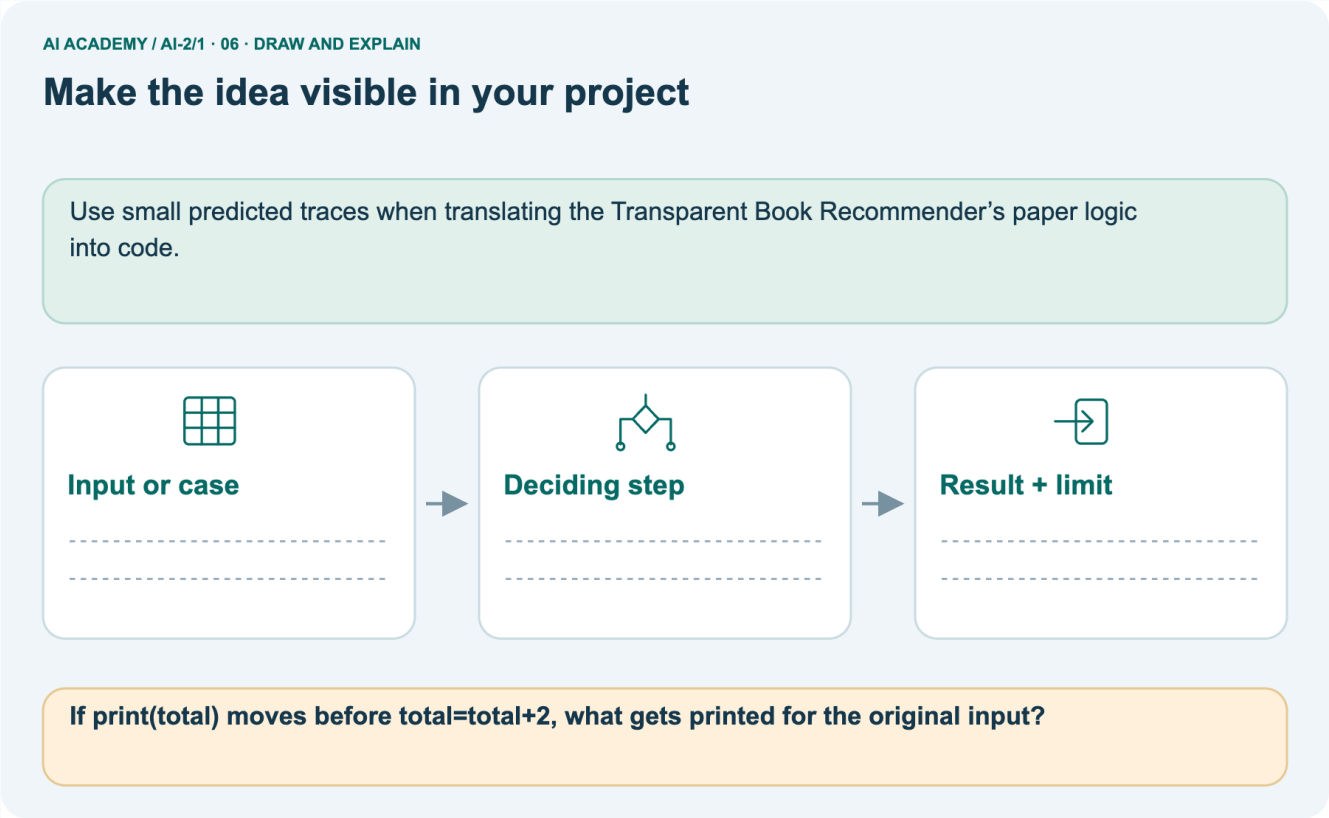
The equal sign is an instruction here, not a claim that 15 and 17 are mathematically equal. Predict **output** before running.

Limit: These statements count supplied books; they do not validate **input** types or infer stock from measurements.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Use small predicted traces when translating the Transparent Book Recommender's paper logic into code. Draw your **input**, deciding step, result and limit.

Supported question: If print(total) moves before total=total+2, what gets printed for the original **input**?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Precision in Algorithms: the case

02. Follow the deciding steps

03. A calculation from paper to output

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.